

From Shellfish to Rice Paddies: Reassessing the Jomon-to-Yayoi Food Transition

Beneath the mudflats of Japan's Pacific coast lie mounds of shell built up, generation after generation, by hands that fed themselves on clams, oysters, and whatever the tide gave up. For more than ten thousand years, the hunter-gatherer communities archaeologists group under the label "Jomon" lived this way, drawing on shellfish, marine and terrestrial mammals, and a wide range of wild plants to build one of the longest-lived non-agricultural traditions in world archaeology. Then, as early as the eleventh century BCE, something changed: communities in the archipelago began cultivating wet rice (*Oryza sativa*), and within a few centuries the crop had spread west to east across most of Japan, eventually giving its name to the following Yayoi period (Leipe et al. 2020). Why did coastal Jomon communities, whose diet had sustained them for millennia, move toward a labor-intensive grain, and what did that shift actually change about their relationship to food? Drawing on archaeobotanical, zooarchaeological, and stable isotope evidence, I argue that the transition was neither sudden nor complete. Rather than replacing foraging outright, coastal communities folded rice into an existing subsistence base over many generations, a pattern archaeologists call "low-level food production" (Smith 2001). What changed was not simply what people ate, but how they related to food itself, as communities moved from an economy built around the immediate, seasonal availability of wild resources toward one increasingly built around the deferred, storable yields of a cultivated grain—a shift set in motion as much by upheaval on the Asian mainland as by anything happening within Japan.

Understanding what changed requires knowing what the Jomon diet looked like before rice ever arrived. The clearest archaeological signature of that diet is the shell midden: a refuse

heap built up over generations from the discarded shells of harvested mollusks. Junko Habu and colleagues' survey of Japanese shell midden archaeology documents thousands of such sites along the Pacific coast, with especially dense concentrations around bays like Matsushima, and traces intensive shellfish collecting back to roughly 8000–7000 BCE, when a radiocarbon-dated oyster shell from the Natsushima site marks the start of large-scale marine exploitation in the Early Jomon period (Habu et al. 2011). Shell middens preserve well and accumulate in seasonal layers, so they let archaeologists reconstruct not just which species people harvested but when. At the Awazu site, alternating bands of freshwater clam and nutshell record repeated spring, summer, and autumn collection episodes, and caloric reconstructions there found that nuts, not shellfish, supplied the largest share of the estimated diet: 52 percent, against 16.7 percent from shellfish. The Isarago midden tells a similar story, where simulation studies estimated that shellfish supplied only about 7 to 9 percent of daily caloric intake (Habu et al. 2011). Shellfish, in other words, mattered to Jomon communities less as a caloric staple than as insurance: a reliable resource during the late winter and early spring, when other foods ran thin, buffering risk rather than dominating the diet. Stable isotope analysis of human bone collagen, a method that measures the ratios of carbon and nitrogen isotopes retained in skeletal tissue to estimate how much of a person's protein came from marine versus terrestrial sources, reinforces this picture. Kusaka and colleagues' isotopic study of Jomon skeletons from two coastal regions found a mixed marine-terrestrial diet whose balance shifted with local resource availability, consistent with terrestrial foods dominating the overall Jomon diet even in coastal, shell-midden-forming communities (Kusaka et al. 2010). A related isotopic study of 123 individuals from the Yoshigo shell mound in Aichi Prefecture, dated to roughly 3200–2800 cal BP, found substantial variation in marine dietary dependence within a single Jomon population, correlated with burial cluster

and, in some clusters, with sex and the presence of ornaments. Jomon subsistence, then, was not only broad-based but socially differentiated well before rice ever arrived (Kiriya and Kusaka 2017). Together, the zooarchaeological and isotopic evidence describes an economy of real flexibility: one that drew on whichever combination of shellfish, fish, terrestrial game, and wild plants was locally most reliable, among people who already ate somewhat different diets from one another.

Pinning down exactly when and how rice cultivation entered this system takes different methods, since carbonized grains and other plant remains survive far more poorly than shell or bone. Paleoethnobotanists lean heavily on impressions left by rice husks and grains in unfired clay, which are preserved permanently once the pottery is fired and can reveal a crop's presence even where the grains themselves have long since rotted away. Endo and Leipe's survey of seed impressions in Jomon and Yayoi pottery from across the archipelago found that rice impressions appear first in southwestern Japan and spread northeastward only gradually over the following centuries, with foxtail millet (*Setaria italica*) and broomcorn millet (*Panicum miliaceum*) serving as earlier, complementary crops in some regions before rice cultivation took over (Endo and Leipe 2022). The method has a real limit, one the authors acknowledge themselves: an impression records only that a plant was present at a site, not how much of a community's diet it supplied, which makes it better for tracing geography and timing than for measuring dietary importance. Chronological precision comes instead from radiocarbon dating performed directly on charred rice grains rather than on associated charcoal, avoiding the risk of dating fuel wood or other material of a different age than the crop itself. Leipe and colleagues' Bayesian modeling of these direct dates found that rice could have appeared as early as the eleventh century BCE in the Central Highlands of Honshu, likely used mainly for ritual purposes among Final Jomon

communities, before turning up in western Japan (northern Kyushu) in the ninth century BCE and then spreading discontinuously across eastern Japan (Leipe et al. 2020). What drove that timing, Leipe and colleagues argue, was less anything happening within Japan than geopolitical upheaval on the Asian mainland. The eastward expansion of the Shang and Zhou dynasties in China, combined with a mid-to-late Holocene decline in summer monsoon precipitation that hurt agricultural yields near the middle Yellow River, pushed early rice-farming populations first onto the Korean peninsula and then, by around 1000 BCE, into Japan (Leipe et al. 2020). Rice's arrival in the Jomon world, understood this way, was not a local innovation or proof of superior productivity. It was the archaeological signature of a chain of migrations triggered by political and climatic pressure thousands of kilometers away.

Even after immigrant rice farmers reached the archipelago, the archaeological record makes clear that rice cultivation did not simply replace Jomon foraging on arrival. Horiuchi and colleagues' combined stable isotope and organic residue analysis of Final Jomon Kamegaoka culture pottery, dated to the centuries just before the conventional Jomon–Yayoi boundary, found chemical signatures consistent with the continued processing of marine and terrestrial foods in cooking vessels, with no clear evidence that cultivated starches had yet displaced these resources at the sites studied (Horiuchi et al. 2015). This fits the broader pattern archaeologist Bruce Smith termed “low-level food production”: a long middle ground between pure foraging and full agricultural dependence, in which cultivated and wild resources are exploited side by side, sometimes for centuries, before agriculture comes to supply the bulk of a community's calories (Smith 2001). Kazunobu Ikeya's ethnoarchaeological analysis of the Jomon-to-Yayoi transition argues that this gradual absorption was possible because incoming rice cultivation was folded into existing patterns of settlement and social organization, rather than requiring wholesale

cultural replacement, drawing on ethnographic analogies with more recent hunter-gatherer groups who selectively adopted cultivation without abandoning foraging (Ikeya 2021). Set beside the isotopic and archaeobotanical evidence discussed above, this comparison across residue chemistry, ethnoarchaeology, and radiocarbon-dated crop remains points to a transition measured in generations rather than years: uneven across regions, and shaped as much by local ecology and existing social structures as by the simple availability of a new crop that migrating farmers brought with them.

So if rice did not immediately replace shellfish and other wild foods, what actually changed? The clearest shift is not in the raw list of foods people ate but in the temporal and social structure of subsistence itself. Shellfish collection, as the seasonal layering and caloric modeling from sites like Awazu and Isarago demonstrate, was organized around short-term, immediate returns: a bed of clams could be harvested within a day of locating it, and the resource replenished on a seasonal cycle largely outside human control (Habu et al. 2011). Rice cultivation demanded something different. Jomon and early Yayoi communities had to invest labor months before any return was realized, through field preparation, planting, weeding, and water management, traces of which survive archaeologically at Yayoi sites as paddy bunds, irrigation canals, and other water-control features. That shift toward delayed-return, storable production carries consequences well beyond diet. A stored rice surplus creates the possibility of unequal accumulation, and control over land and water becomes a new axis of social differentiation in a way that a naturally replenishing, widely accessible shellfish bed simply could not. Jomon society was not perfectly egalitarian to begin with: Kiriama and Kusaka (2017) documented variation in marine dietary dependence by burial cluster, sex, and mortuary ornamentation even within the Late and Final Jomon period at Yoshigo. But the later addition of

a storable, labor-intensive staple, introduced alongside new immigrant populations, plausibly accelerated those existing tendencies toward social differentiation, a pattern archaeologists have observed in agricultural transitions across many other world regions.

The archaeological evidence for the Jomon-to-Yayoi transition resists any simple story of abrupt replacement. Zooarchaeological and isotopic evidence show that coastal Jomon communities relied on a broad, seasonally structured diet in which shellfish buffered risk rather than supplying most calories. Archaeobotanical and radiocarbon evidence show that rice arrived first as a ritual crop in the Central Highlands before spreading economically from northern Kyushu, propelled by migrations tied to political and climatic disruption on the Asian mainland. And residue, isotopic, and ethnoarchaeological evidence together show that the resulting subsistence systems coexisted, sometimes for centuries, before agriculture finally came to predominate. What changed most fundamentally, then, was not simply the presence of rice on the table but the underlying logic of subsistence itself: a move from resources that were immediately available and broadly accessible toward a staple that demanded sustained labor investment and opened new possibilities for storage, surplus, and social inequality. Additional isotopic sampling from underrepresented regions of eastern Japan, and further direct radiocarbon dating of early rice remains, would help clarify how evenly this transition actually proceeded across the archipelago. Even so, the methodologically diverse evidence assembled here already makes clear that the shift from shellfish to rice was not a single, local decision. It was the plausible outcome of a gradual, geographically variable renegotiation of how people related to food, one ultimately set in motion by forces well beyond the Jomon world itself.

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